

Using gdb:

```
"Auto-loading safe path" section in the GDB manual. E.g., run from the shell:
info "(gdb)Auto-loading safe path"
(gdb) target remote localhost:26000
Remote debugging using localhost:26000
0x0000000000001000 in ?? ()
(gdb) b syscall
Breakpoint 1 at 0x80001ce8: file kernel/syscall.c, line 133.
(gdb) c
Continuing.
[Switching to Thread 1.2]

Thread 2 hit Breakpoint 1, syscall () at kernel/syscall.c:133
133 {
(gdb) n
135     struct proc *p = myproc();
(gdb) n
137     num = p->trapframe->a7;
(gdb) n
138     if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
(gdb) p /x *p
$1 = {lock = {locked = 0x0, name = 0x80007130, cpu = 0x0}, state = 0x4, chan = 0x0, killed = 0x0, xstate = 0x0,
pid = 0x1, parent = 0x0, kstack = 0x3fffffff000, sz = 0x4000, pagetable = 0x87f52000, trapframe = 0x87f56000,
context = {ra = 0x80001306, sp = 0x3fffffffdc50, s0 = 0x3fffffffdc80, s1 = 0x8000a630, s2 = 0x8000a200, s3 = 0x0,
s4 = 0x80015b98, s5 = 0x3, s6 = 0x8001b2d0, s7 = 0x0, s8 = 0x8001b3f8, s9 = 0x180, s10 = 0xb0, s11 = 0x2},
ofile = {0x0 <repeats 16 times>}, cwd = 0x80018740, name = {0x69, 0x6e, 0x69, 0x74, 0x0 <repeats 12 times>}}
(gdb) p /x p->trapframe->a7
$2 = 0xf
(gdb) p /x $sstatus
$3 = 0x200000022
^C
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=
medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-m
emset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-
fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/syscall.o kernel/syscall.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/st
ring.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/
sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysf
ile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.
o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/
_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/
_forphan user/_dorphan user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 973 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
scause=0xd sepc=0x80001cf8 stval=0x0
panic: kerneltrap
```



```
suffi@PC: ~/xv6-labs-2025
emset -fno-builtin-memmove -fno-builtin-memcpy -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-
fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/secret.o user/secret.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/ulib.o user/usys.o user/pr
intf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_secret > user/secret.asm
riscv64-linux-gnu-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/
_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/
_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmata 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1009 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ ./grade-lab-syscall sandbox_mask
exec ./grade-lab-syscall failed
$ QEMU: Terminated
suffi@PC:~/xv6-labs-2025$ ./grade-lab-syscall sandbox_mask
make: 'kernel/kernel' is up to date.
== Test sandbox_mask == sandbox_mask: OK (1.2s)
suffi@PC:~/xv6-labs-2025$
```

Sandbox with allowed path names:

```
suffi@PC: ~/xv6-labs-2025
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-
fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/secret.o user/secret.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/ulib.o user/usys.o user/pr
intf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_secret > user/secret.asm
riscv64-linux-gnu-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/
_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/
_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmata 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1009 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(kaashoek,rtm@mit.edu). The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ ./grade-lab-syscall sandbox_mask
```

```

suffi@PC: ~/xv6-labs-2025
sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysf
ile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.
o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/suffi/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: FAIL
    Cannot read answers-syscall.txt
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (4.8s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.9s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.3s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.4s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.2s)
== Test attack ==

```

Attack task:

```

medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-buil
emset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-
-fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-bui
fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/
ring.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o ke
sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/
ile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/
o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test attack == attack: OK (3.8s)
    (Old xv6.out.attack failure log removed)
suffi@PC:~/xv6-labs-2025$ |

```

All tasks done:

```

o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test attack == attack: OK (3.8s)
    (Old xv6.out.attack failure log removed)
suffi@PC:~/xv6-labs-2025$ git add user/attack.c
suffi@PC:~/xv6-labs-2025$ git commit -m "Completed Attack xv6 vulnerability task"
[problemSet2 36d1c3e] Completed Attack xv6 vulnerability task
    1 file changed, 50 insertions(+), 4 deletions(-)
suffi@PC:~/xv6-labs-2025$ git push
Username for 'https://github.com': suffiyan-asghar
Password for 'https://suffiyan-asghar@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 984 bytes | 328.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/suffiyan-asghar/xv6-labs-2025.git
    500d734..36d1c3e  problemSet2 -> problemSet2
suffi@PC:~/xv6-labs-2025$

```